

A provisional Kibble–Zurek cosmology programme note for TECT
(NOT a canonical GAP-4 closure paper. The canonical F-GAP4 shell per Math300 / Math311 is a defect-mass verdict window $\mu_{\text{defect}} \in [10^{13}, 10^{17}]$ GeV, NOT an SKA/IPTA non-detection condition. The predicted Ω_{GW} peak frequency in the present draft is internally inconsistent with PTA bands. External use FORBIDDEN at the present canonical tier per Math314-AddD audit.)

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This document is a PROVISIONAL Kibble–Zurek (KZ) cosmology programme note for the Topological Energy Condensate Theory (TECT). The KZ mechanism applied to the BCC topological condensate freeze-out generates a spectrum of topological defects (candidate monopoles, vortices, BCC domain walls) whose gravitational-wave signature is sharply peaked at a frequency corresponding to the condensate shell wavenumber. The original draft proposed a PROVISIONAL Ω_{GW} peak at $f \sim 10^{-16}$ Hz (incorrectly labelled “pulsar-timing array band”) and re-defined the F-GAP4 gate as an SKA/IPTA non-detection condition.

Audit-flag (AUDIT-2026-05-02-Wave7, Math314-AddD): both elements are inconsistent with current canonical bookkeeping and with standard GW-observation phenomenology: (a) the proposed $f \sim 10^{-16}$ Hz peak is below ALL existing GW observational bands (PTA: nHz $\sim 10^{-9}$ Hz; LISA: mHz; LIGO: kHz). The original “PTA band” label is incorrect; the paper later cites SKA/IPTA sensitivity at $\sim 10^{-9}$ Hz, contradicting its own peak-frequency prediction. The Ω_{GW} peak frequency must be re-derived from first principles before external citation. (b) the canonical F-GAP4 shell per Math300 / Math311 is a defect-mass verdict window $\mu_{\text{defect}} \in [10^{13}, 10^{17}]$ GeV with solver / grid certificates, NOT an SKA/IPTA non-detection condition. Redefining F-GAP4 within this paper would change the canonical gate without going through the canonical-source-hierarchy update process. (c) “Paper 16 completes the GAP-4 closure” wording is incompatible with the current GAP-4 = T3 / Stage-2 composite = T3 canonical tier.

External use of the present draft is FORBIDDEN at the present canonical tier; the document is preserved as a candidate KZ-cosmology programme note pending (i) re-derivation of the Ω_{GW} peak frequency, (ii) honest separation between this candidate cosmological prediction and the canonical F-GAP4 defect-mass verdict shell, and (iii) demotion of the closure claim to a candidate-prediction framing.

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I. INTRODUCTION — PROVISIONAL KZ-COSMOLOGY PROGRAMME, NOT A CURRENT CANONICAL GAP-4 CLOSURE PAPER

Kibble–Zurek (KZ) formation of topological defects is a universal mechanism in phase transitions: when a system is cooled through a critical point faster than the microphysical relaxation timescale, causally disconnected regions freeze out in different ground states and topological defects appear at the boundaries between them. In standard cosmology the mechanism is invoked in axion physics, in beyond-SM monopole and cosmic-string scenarios, and in the freeze-out of light-pseudo-scalar fluctuations.

TECT proposes a concrete realisation in the BCC topological condensate: the condensate undergoes a phase transition as the universe cools below the Brazovskii instability scale $T_{\text{crit}} \sim 10^{16}$ GeV, and the KZ mechanism generates a spectrum of candidate defects (monopoles, vortices, BCC domain walls) with characteristic wavenumber $\sim q_0 \sim \ell_P^{-1}$. These defects can in principle radiate gravitational waves, and the resulting stochastic background $\Omega_{\text{GW}}(f)$ is the natural target of any “cosmological observables” programme for TECT.

Honest scope of this paper. The present document is a *provisional* KZ-cosmology programme note. It is *not* a current canonical GAP-4 closure paper. Two structural points must be made up-front because earlier drafts mis-stated them:

- The naive estimate of the peak frequency of the Ω_{GW} spectrum at the TECT operating point sits far below all existing GW observational bands, including the pulsar-timing-array (PTA) band ($f \sim 10^{-9}$ Hz). The peak frequency derived in §II C below is *not* in the PTA band; the earlier “directly observable via PTA” wording is *withdrawn* (Remark 1).
- The canonical F-GAP4 falsification gate per Math300 / Math311 is a *defect-mass verdict-window shell*, $\mu_{\text{defect}} \in [10^{13}, 10^{17}]$ GeV with PASS / FAIL / DEFER on solver and grid certificates. F-GAP4 is *not* an SKA / IPTA non-detection condition; the SKA / IPTA gate of an earlier draft was a paper-internal redefinition and is *withdrawn* (Remark 3).

Both withdrawals are recorded in this paper as honest scope clarifications, and the F-GAP4 section below uses the canonical defect-mass verdict-window shell, not the SKA/IPTA wording.

II. THE KIBBLE-ZUREK DEFECT SPECTRUM

A. Cooling rate and freeze-out time

The universe's expansion rate (Hubble parameter) during TECT-relevant epochs is:

$$H(T) = \frac{\dot{a}}{a} \sim \frac{T^2}{M_{\text{Planck}}} \quad (1)$$

for a radiation-dominated universe. The BCC condensate critical temperature is (from the Brazovskii instability):

$$T_{\text{crit}} \sim M_X, \quad (2)$$

where M_X is the characteristic shell energy (related to the wavenumber via $M_X \sim q_0 c_T$).

The freeze-out temperature is determined by equating the relaxation timescale $\tau_{\text{relax}} \sim H^{-1}(T_{\text{freeze}})$:

$$\tau_{\text{relax}}(T_{\text{freeze}}) \sim (T_{\text{crit}} - T_{\text{freeze}})^{1/2} \cdot |\dot{T}|^{-1}, \quad (3)$$

where $|\dot{T}| \sim HT$ is the cooling rate. Solving for T_{freeze} yields (Math196):

$$T_{\text{freeze}} \sim T_{\text{crit}} \left(1 - \left(\frac{T_{\text{crit}}}{M_{\text{Planck}}} \right)^{1/4} \right). \quad (4)$$

B. Defect density and correlation length

The KZ prediction for defect density is:

$$n_{\text{defect}} \sim \xi_{\text{freeze}}^{-3}, \quad \xi_{\text{freeze}} \sim (|\dot{T}|/T_{\text{crit}}^2)^{-1/4}. \quad (5)$$

Using the cosmological cooling rate $|\dot{T}| \sim H(T_{\text{freeze}})T_{\text{freeze}}$:

$$\xi_{\text{freeze}} \sim \left(\frac{M_{\text{Planck}}}{T_{\text{freeze}}^2} \right)^{1/4}. \quad (6)$$

For the BCC case (Math159), the freeze-out correlation length at the TECT operating point is:

$$\xi_{\text{freeze}} \sim (10^3 \text{ to } 10^4) \ell_P, \quad (7)$$

yielding a defect number density $n_{\text{defect}} \sim (10^{-13} \text{ to } 10^{-15}) \ell_P^{-3}$ at that epoch.

C. Naive peak-frequency estimate from defect dynamics

Topological defects of characteristic size ξ_{freeze} oscillate and radiate gravitational waves at a frequency set by the inverse oscillation timescale,

$$f_{\text{GW}}^{(\text{rest})} \sim \frac{1}{2\pi \tau_{\text{osc}}}, \quad \tau_{\text{osc}} \sim \xi_{\text{freeze}}/cT. \quad (8)$$

In the observer frame the frequency is redshifted by the inverse-temperature factor,

$$f_{\text{obs}} = f_{\text{GW}}^{(\text{rest})} \frac{a(T_{\text{freeze}})}{a_0} = f_{\text{GW}}^{(\text{rest})} \frac{T_0}{T_{\text{freeze}}}, \quad (9)$$

with $T_0 \approx 2.7\text{ K}$ the present CMB temperature.

For the BCC freeze-out at $T_{\text{freeze}} \sim M_X \sim 10^{16}\text{ GeV}$, the naive insertion of $\xi_{\text{freeze}} \sim (10^3\text{--}10^4)\ell_P$ from §II B gives, to within an order of magnitude,

$$f_{\text{obs}}^{\text{naive}} \sim 10^{-16} \text{ to } 10^{-15} \text{ Hz}. \quad (10)$$

Remark 1 (The estimate (10) is NOT in the PTA band, and is below ALL existing GW observational bands). The PTA band of the SKA / IPTA / NANOGrav programmes spans $f \sim 10^{-9}$ to 10^{-7} Hz. The naive estimate (10) sits at $f \sim 10^{-16}$ Hz, that is, *seven orders of magnitude below the PTA lower edge*. It is also far below the LISA band ($\sim$ mHz) and the LIGO band ($\sim$ kHz). The earlier-draft labelling of (10) as “in the PTA band” is *factually incorrect* and is withdrawn here. The estimate (10) is closer to the Hubble-scale band probed by CMB polarisation B-mode searches; even there, the predicted amplitude $\Omega_{\text{GW}}h^2 \sim 10^{-15}$ of §III is well below current and projected sensitivities. The estimate (10) is therefore *not currently observable in any GW band*; this is a programme limitation, not a closure prediction.

III. PROVISIONAL Ω_{GW} -SPECTRUM ESTIMATE (PROGRAMME-LEVEL ONLY)

The gravitational-wave energy density per logarithmic frequency interval is

$$\Omega_{\text{GW}}(f) h^2 = \int_0^\infty dt \int_{\text{modes}} \frac{d\rho_{\text{GW}}}{df}(t, f). \quad (11)$$

For KZ-formed defects the spectrum is peaked near the characteristic frequency $f_{\text{peak}}^{\text{naive}}$ of §II C with a bandwidth $\Delta \log f \sim 1$ (defect-type dependent). Math172 [4] computes the detailed spectrum at the TECT operating point $(q_0, M_X, \lambda, \gamma, Y) = (14.32, 10^{16}\text{ GeV}, -0.0394, -0.0188, 280.2)$ and gives the *provisional* estimate

$$\Omega_{\text{GW}}(f_{\text{peak}}) h^2 \sim 10^{-15} \quad \text{at } f \sim 10^{-16} \text{ Hz}. \quad (12)$$

Remark 2 (The estimate (12) is provisional and NOT a closure prediction). (12) is reported as a programme-level estimate only. It is sensitive to (i) the operating point (particularly the coupling λ , which controls the defect core size), (ii) the four-defect-repair verdict pending in Math218-AddA, and (iii) the freeze-out cooling-rate normalisation Math196 / Math159. The estimate is *not* part of any current canonical falsification gate; per Remark 1, the predicted peak frequency is below all existing GW observational bands and cannot be tested by any current or near-future Ω_{GW} measurement.

IV. F-GAP4: THE CANONICAL DEFECT-MASS VERDICT-WINDOW SHELL (NOT AN SKA/IPTA NON-DETECTION CONDITION)

Remark 3 (F-GAP4 is the Math300/Math311 defect-mass verdict shell, NOT an SKA / IPTA non-detection condition). The canonical F-GAP4 falsification gate per Math300 / Math311 is the defect-mass verdict-window shell

$$\boxed{\text{F-GAP4 : } \mu_{\text{defect}} \in [10^{13} \text{ GeV}, 10^{17} \text{ GeV}]} \quad (13)$$

extracted from the lowest non-trivial eigenvalue $\lambda_{\min}(\square_E)$ of the candidate transverse Box operator $\square_E$ on the defect manifold, with PASS / FAIL / DEFER on the canonical solver and grid certificates registered in Math300 and the verdict-shell consumption tracked in Math311. F-GAP4 is *not* an SKA / IPTA non-detection condition; the SKA / IPTA wording of an earlier draft was a paper-internal redefinition that conflicted with the canonical-source hierarchy (the internal canonical-source hierarchy). The earlier wording is *withdrawn*.

a. Status of the four current GW observational bands relative to the present provisional estimate. The present estimate (12) is provided for completeness only; *none* of the current GW observational bands overlaps with the predicted peak frequency:

- PTA band (SKA / IPTA / NANOGrav): $f \sim 10^{-9} \text{ Hz}$ — *seven orders of magnitude above* the predicted peak.
- LISA band: $f \sim 10^{-3} \text{ Hz}$ — *thirteen orders of magnitude above* the predicted peak.
- LIGO / Virgo band: $f \sim 10^2 \text{ Hz}$ — *eighteen orders of magnitude above*.
- CMB B-mode polarisation (Hubble-scale): the closest band in frequency, but the predicted amplitude $\Omega_{\text{GW}} h^2 \sim 10^{-15}$ is well below current and projected B-mode sensitivities.

The earlier-draft assertion that the present prediction is “directly observable via pulsar-timing arrays” is therefore *factually wrong* and is withdrawn together with the F-GAP4 SKA / IPTA redefinition.

V. RELATION TO STAGE-2 AND OTHER PILLARS

Honest status. The present provisional KZ-cosmology programme note does *not* complete a GAP-4 closure. The current canonical GAP-4 tier per Math307 / Math310 is **T3 proof sketch**, with promotion gated by the canonical defect-mass verdict-window shell (13) of Math300 / Math311 (PASS / FAIL / DEFER on solver and grid certificates), *not* by any SKA / IPTA non-detection. The earlier-draft wording “Paper 16 completes the GAP-4 closure” is incompatible with the current canonical Stage-2 composite tier $\min(T6, T6, T4, T3) = T3$ and is *withdrawn*.

a. What an F-GAP4 PASS would entail. A PASS on the canonical defect-mass verdict-window shell (13) would advance Pillar 4 sub-task 3 (the defect-mass prediction) to T6 PROVED CONDITIONAL and would simultaneously support the BCC freeze-out cooling profile of Math159 / Math196 at $T_{\text{freeze}} \sim 10^{16}$ GeV. It would *not* by itself certify the provisional Ω_{GW} estimate of (12), which is unobservable in the near term per Remark 1.

b. What an F-GAP4 FAIL would entail. A FAIL on (13) would require (i) revision of the KZ rescope of Math159 / Math196 (scope narrowing or T -scaling modification); (ii) re-evaluation of the defect-mass prediction in Pillar 4 sub-task 3; and (iii) possible reversion of Stage-2 composite-tier claims dependent on the defect-mass shell.

VI. CONCLUSION — PROVISIONAL PROGRAMME NOTE, NOT A CLOSURE PAPER

The KZ defect-formation mechanism in TECT generates a candidate stochastic gravitational-wave background whose order-of-magnitude peak-frequency estimate sits at $f \sim 10^{-16}$ Hz — seven or more orders of magnitude below all current GW observational bands (Remark 1). The accompanying amplitude estimate $\Omega_{\text{GW}} h^2 \sim 10^{-15}$ is provided only as a programme-level estimate and is *not* part of any current canonical falsification gate (Remark 2). The canonical F-GAP4 gate is the defect-mass verdict-window shell $\mu_{\text{defect}} \in [10^{13}, 10^{17}]$ GeV of (13), with PASS / FAIL / DEFER on solver / grid certificates per Math300 / Math311; it is *not* an SKA / IPTA non-detection condition (Remark 3).

The present document is therefore preserved as a *provisional KZ-cosmology programme note*

pending (i) re-derivation of the Ω_{GW} peak frequency from a more careful defect- dynamics calculation; (ii) honest separation between the cosmological prediction and the canonical defect-mass verdict-window F-GAP4 gate; and (iii) the four-defect-repair verdict tracked in Math218-AddA (deadline 2026-06-15). External use is FORBIDDEN at the present canonical tier per Math314-AddD audit.

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- [1] J. Lee, *Topological Energy Condensate Theory (TECT) Repository*, <https://github.com/...> (2026).
 - [2] J. Lee, “Round V1–V5 comprehensive audit verdict,” *Math156*, 2026-04-25.
 - [3] J. Lee, “KZ rescope from inflation,” *Math159*, 2026-04-25.
 - [4] J. Lee, “PROVISIONAL prediction $\Omega_{\text{GW}} \sim 10^{-15}$ at PTA-band frequency,” *Math172*, 2026-04-25.
 - [5] J. Lee, “KZ rate Friedmann coupling,” *Math196*, 2026-05-01.
 - [6] J. Lee, “E3’ cosmological realization four-defect repair,” *Math218-AddA*, 2026-04-29.